

DESCRIPTIVE STATISTICS

STATISTICS

Statistics is a branch of mathematics that deals with collecting, analyzing, interpreting, presenting, and organizing data. It provides tools for understanding variability, making informed decisions, and drawing conclusions based on data.

Importance of Statistics in Data Science

- **DATA CLEANING**

Collect and structure data for machine learning algorithms, eliminate irrelevant data and redundancies.

- **DATA ANALYSIS**

Analyze and assess results from data usage and helps to determine the next steps for data improvement.

- **STATISTICAL METHODS FOR UNDERSTANDING DATA**

Calculate numerical values such as variance, mean, probability and distribution for understanding the data.

There are two main **types of statistics**:

1. **DESCRIPTIVE STATISTICS**

2. **INFERENTIAL STATISTICS**

Descriptive analysis in statistics focuses on the methods of describing the characteristics of a dataset. It helps to summarize, explore and understand data, making conclusions about data to make rational decisions. Descriptive statistics are the first step in data analysis and do not involve making inferences or generalizations about the population.

Here are the most important topics, briefly explained:

1. **Measures of Central Tendency**

- **Mean:** Average value of the dataset.
- **Median:** Middle value when data is sorted.
- **Mode:** Most frequently occurring value.

2. **Measures of Dispersion (Variability)**

- **Range:** Difference between max and min values.
- **Variance:** Average squared deviation from the mean.
- **Standard Deviation:** Square root of the variance, shows data spread.
- **Interquartile Range (IQR):** Range between the 25th and 75th percentiles.

3. Shape of Distribution

- **Normal Distribution**
- **Skewness:** Measures asymmetry (left or right skewed).
- **Kurtosis:** Measures peakedness or flatness of the distribution.

4. Data Visualization

- **Histograms:** Show frequency distribution.
- **Box Plots:** Show spread and detect outliers.
- **Bar Charts & Pie Charts:** For categorical data summaries.
- **Scatter Plots:** Show relationships between two variables.

Uses of Descriptive Statistics

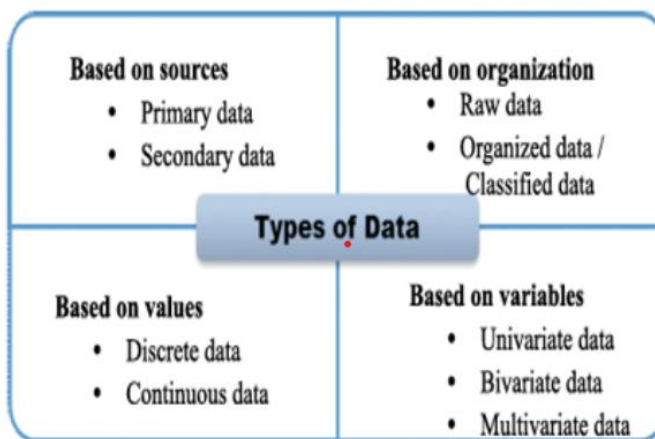
- **Data Exploration:** Identify patterns, trends, and anomalies.
- **Summarization:** Simplify complex datasets.
- **Communication:** Provide an overview for reports or presentations.

Inferential Statistics Makes predictions or inferences about a population based on a sample of data. It goes beyond merely describing the data to making conclusions and decisions based on the analysis.

This involves:

- **Hypothesis testing.**
- **Estimation of parameters.**
- **Confidence Intervals**

Types of data



Data Based on Source (Primary/Secondary)

Based upon the sources of collection, data can be primary or secondary.

If data is collected directly from the fields or from individuals (surveys, observations, case study) the data is called **Primary data**. If it is taken from already collected data then such data is called **Secondary data**.

Ex: If teacher collects the data by measuring the height of each student, then data so collected is called primary data.

If teacher refer the medical records of the students to note their heights then such data is called secondary data.

Data based on Variables. (Univariate, Bivariate, Multivariate data)

Univariate data involves only **one variable**. For example: height of all students in the class.

Ex: Temperature of Delhi during last seven days (in degree celcius) is univariate data.

Temperature(°C)	40	41	37	38	34	38	36	39	40
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Bivariate data involves two different variables. The analysis is done to find out the connection between these two variables, mainly the causes and relationships.

Ex: Demand curve correctly explains the bivariate data where quantity demanded for a goods has an inverse relationship with the price (keeping other factors constant like income)



Other examples for bi variate data is Distance – Time graph, calorie intake – weight graph, which compares two variables.

Multi Variate data involves two or more variables where analysis is done to understand the interactions between different fields of the data.

Ex: Your grade in class XI depends on the performance in various subjects (i.e. multiple variables) you have opted.

Profit of a company depends on the various statistics like cost of raw material, labour, marketing expenditure, sales etc.

Data Based on Organization (Raw/ organized data)

When we first collect the data, it is mostly unorganized. This first hand collected data is called Raw Data (Primary data). When we organize the raw data in order to draw any meaningful conclusion out of them, it is called organized/ classified data.

Ex: ODI runs secured by Virat Kohli in the year 2019 given below is raw data.

3	104	46	45	43	60	44	116	123	7
20	18	82	77	67	72	66	26	34	1
120	114	4	0	85	23	25	64	78	20

This data needs to be organized in a proper order before any systematic statistical analysis is undertaken.

Data Based on values (Discrete/ Continuous data)

Discrete data refers to information that can only take specific, separate values and is typically countable, such as the number of patients, hospital rooms, or phone calls.

In contrast, continuous data can take any value within a given range and is measurable rather than countable. This type of data includes variables like weight, temperature, blood pressure(90 – 120), or waiting time (1pm – 3pm)

While discrete data deals with distinct quantities, continuous data allows for infinite possibilities within a range.

STAGES IN STATISTICAL INVESTIGATION

1. Collection of data
2. Organization of data
3. Presentation of data (tabular or graphical presentation)
4. Analysis of data
5. Inference of data

After collection of data, the **data can be organized** in classes or categories which supports in the **presentation and analysis** of data. So data can be organized as:

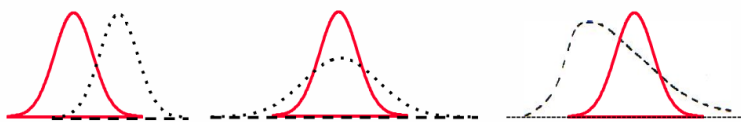
- **Raw Data**
- **Ungrouped frequency distribution (Discrete data)**
- **Grouped frequency distribution (continuous data)**

The classified data **can be presented graphically** as Bar Chart, Pie Chart, Pictographs, Histogram, Frequency Polygon.

Descriptive Statistics

The following measures are used to describe a data set:

- ❑ Measures of position (also referred to as central tendency or location measures).
- ❑ Measures of spread (also referred to as variability or dispersion measures).
- ❑ Measures of shape.

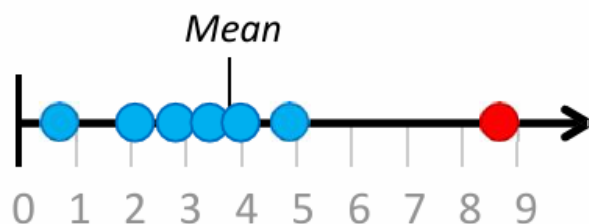


MEASURE OF POSITION

Measures Of Central Tendency

The **central tendency measure** is the center or middle value of a set of data values, that represent the data (around which data values tend to cluster).

- **MEAN (Arithmetic, Geometric, Harmonic):** The average value, which is sensitive to outliers. It is most commonly used statistic of position. It is easy to understand and calculate. It works well when distribution is symmetric and there are no outliers.



1.Simple Arithmetic Mean for RAW data

Add all the observations together and obtain the total $\sum X$

Divide the total by number of observations

$$\bar{X} = \frac{X_1 + X_2 + X_3 \dots + X_n}{N}$$

$$\bar{X} = \frac{\sum X}{N}$$

1.Simple Arithmetic Mean for ungrouped data (discrete series) / grouped data (continuous series)

*** Direct Method...**

Multiply each value by its frequency to obtain the total $\sum fx$. Divide the total by number of observation that is the total frequency.

$$\bar{X} = \frac{\sum fX}{N}$$

***Shortcut method - Using Assumed mean**

Take an assumed mean and denote it by A

Take the deviations d of the variable X from the assumed mean ($d=x-a$)

Multiply this deviation by respective frequency and take the total $\sum fd$

Apply the formula

$$\bar{X} = A + \frac{\sum fd}{N}$$

Example: Obtain the mean of the following number

2, 7, 8, 2, 7, 3, 7

Solution:

X_i	f_i	$X_i f_i$
2	2	4
3	1	3
7	3	21
8	1	8
Total	7	36

$$\bar{X} = \frac{\sum_{i=1}^4 f_i X_i}{\sum_{i=1}^4 f_i} = \frac{36}{7} = 5.15$$

***Step deviation method- Using Assumed mean**

Take an assumed mean and denote it by A

Take the deviations d of the variable X from the assumed mean

Find $u = d/i$ (where I is the class size or difference between class marks)

Apply the formula

$$\text{MEAN} = a + \frac{\sum fu}{\sum f} \times i \quad \text{where } u = x - a/i \quad \text{where } i = \text{class interval}$$

For continuous data find x by finding mean of upper limit and lower limit and follow the same above steps. Also the distribution must be exclusive not inclusive.

Ques: From the following data of marks obtained by students, calculate arithmetic mean

Marks	20	30	40	50	60	70
# students	8	12	20	10	6	4

Ques: Data regarding the weights of students of class X of a school is given.
Compute the mean (ans: 57.24)

Weight	50-52	52-54	54-56	56-58	58-60	60-62	62-64
No. of students	18	21	17	28	16	35	15

Ques: Find arithmetic mean for the following data (ans: 36.36)

Class	25-29	30-34	35-39	40-44	45-49	50-54	55-59
Frequency	14	22	16	6	5	3	4

Ques: If mean of the frequency distribution is 54, find p (p= 11)

Class	0-20	20-40	40-60	60-80	80-100
Frequency	7	P	10	9	13

Ques: The daily expenditure of 100 families are given below. Calculate f_1 and f_2 if the mean daily expenditure is Rs.188 ($f_1= 50$, $f_2=15$)

Expenditure	140-160	160-180	180-200	200-220	220-240
No. of families	5	25	F_1	F_2	5

Geometric Mean

Multiply all the values and get the result

Get the square root to the Nth power to find the geometric mean

$$\bar{X} = \sqrt[N]{x_1 * x_2 * \dots * x_n}$$

Harmonic Mean

Get reciprocal of each number and add together. Divide the number of values by the total calculated earlier

$$\bar{X} = \frac{N}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}$$

Ques:

A cyclist pedals from his house to his college at speed of 10 km/hr and back from the college to his house at 15 km/hr. Find the average speed.

Solution: Here the distance is constant

→ The simple H.M is appropriate for this problem.

$X_1 = 10 \text{ km/hr}$

$X_2 = 15 \text{ km/hr}$

$$\text{H.M} = \frac{2}{\frac{1}{10} + \frac{1}{15}} = 12 \text{ km/hr}$$

Weighted Mean

Multiply every value with corresponding weight. Add the values together. Divide the total by sum of all the weights

$$\bar{X} = \frac{\sum W_i X_i}{W_1 + W_2 + \dots + W_n}$$

Weighted mean

$$\bar{x} = \frac{n_1 x_1 + n_2 x_2}{n_1 + n_2}$$

combined mean

Ques: In a class there are 30 females and 70 males. If females averaged 60 in an examination, and boys averaged 72, find the mean of the entire class.

Females

$$\bar{X}_1 = 60$$

$$n_1 = 30$$

Males

$$\bar{X}_2 = 72$$

$$n_2 = 70$$

$$\bar{X}_c = \frac{\bar{X}_1 n_1 + \bar{X}_2 n_2}{n_1 + n_2} = \frac{\sum_{i=1}^2 \bar{X}_i n_i}{\sum_{i=1}^2 n_i}$$
$$\Rightarrow \bar{X}_c = \frac{30(60) + 70(72)}{30 + 70} = \frac{6840}{100} = 68.40$$

Ques:

A student obtained the following percentage in an examination:

English 60, Biology 75, Mathematics 63, Physics 59, and chemistry 55. Find the student's weighted arithmetic mean if weights 1, 2, 1, 3, 3 respectively are allotted to the subjects.

$$\bar{X}_w = \frac{\sum_{i=1}^5 x_i w_i}{\sum_{i=1}^5 w_i} = \frac{60*1 + 75*2 + 63*1 + 59*3 + 55*3}{1 + 2 + 1 + 3 + 3} = \frac{615}{10} = 61.5$$

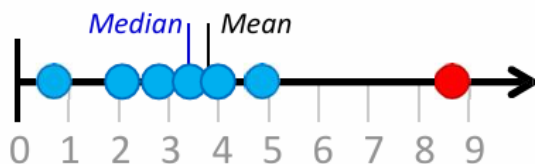
Trimmed Mean:

A trimmed mean is a measure of central tendency, calculated by removing a specific percentage of the smallest and largest values from a dataset before calculating the mean of the remaining data. This approach reduces the impact of outliers.

A percentage (5%, 10%, 20% ..etc) is trimmed from each end of the data, usually arranged in ascending order. **Trimmed mean is used in case of outliers.**

MEDIAN:

The middle value when the data is ordered, which divides the dataset into two halves. It is robust to outliers and skewed data. **It is used when data is nonsymmetrical.**



MEDIAN FOR RAW DATA

Arrange data in Ascending order. Find n

If n is odd, median = $\frac{N+1}{2}$ th term

If n is even, median = $\frac{1}{2} \left[\frac{N}{2} \text{ th term} + \left(\frac{N}{2} + 1 \right) \text{ th term} \right]$

MEDIAN FOR DISCRETE DATA

The value corresponding to the cumulative frequency which is just greater than the $(N+1)/2$, where N is the number of terms.

MEDIAN FOR GROUPED DATA

Median for grouped data is calculated by formula:

$$\text{Median} = l + \left(\frac{\frac{N}{2} - C}{f} \right) \times h$$

Where:

- l = lower boundary of the median class
- N = total number of observations (sum of all frequencies)
- C = cumulative frequency of the class before the median class
- f = frequency of the median class
- h = width (class size) of the median class

REMARK: The median class is the class with the smallest cumulative frequency, greater than $\frac{N}{2}$

Ques: For the following set of data compute the median. (ans: 47.57)

Class Interval	Frequency
0 - 20	6
20 - 40	20
40 - 60	37
60 - 80	10
80 - 100	7

MODE

Mode is a value which occurs most frequently in a set of values, and is useful for categorical data

The mode may not exist and even if it does exist, it may not be unique.

In case of discrete distribution the value having the maximum frequency is the modal value.

Examples:

1. Find the mode of 5, 3, 5, 8, 9

Mode = 5

2. Find the mode of 8, 9, 9, 7, 8, 2, and 5.

It is a bimodal Data: 8 and 9

3. Find the mode of 4, 12, 3, 6, and 7.

No mode for this data.

Mode for Grouped Data

If data are given in the shape of continuous frequency distribution, the mode is defined as:

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

The class interval with maximum frequency is the modal class

Where

l = lower limit of the modal class,

h = class size (assuming class size to be equal)

f_1 = frequency of the modal class,

f_0 = frequency of the class preceding the modal class,

f_2 = frequency of the class succeeding the modal class.

Ques:

Following is the distribution of the size of certain farms selected at random from a district.

Calculate the mode of the distribution

Size of farms	No. of farms
5-15	8
15-25	12
25-35	17
35-45	29
45-55	31
55-65	5
65-75	3

Solutions:

45 – 55 is the modal class, since it is a class with the highest frequency.

$$L_{mo} = 45$$

$$w = 10$$

$$\Delta_1 = f_{mo} - f_1 = 2$$

$$\Delta_2 = f_{mo} - f_2 = 26$$

$$f_{mo} = 31$$

$$f_1 = 29$$

$$f_2 = 5$$

$$\begin{aligned}\Rightarrow \hat{X} &= 45 + 10 \left(\frac{2}{2 + 26} \right) \\ &= 45.71\end{aligned}$$

NOTE: For uni-modal data, there exist a relation between mean, median, mode

$$\text{MODE} = 3 \text{ MEDIAN} - 2 \text{ MEAN}$$

When to replace missing values with mean or median?

Replace missing values with median for most cases, especially skewed data or outliers.
Use mean only for symmetric, normal distributions (Gaussian distribution)

Why impute with median?

Impute missing values with the median for its **robustness against outliers** and **skewed distributions**. It preserves the data's central tendency better than the mean.

Data with outliers that could significantly affect the mean.

Method	Suitable For	Why Not for Categorical?
Mode	Categorical (nominal/ordinal)	Matches existing categories; no numeric assumption
Median	Numerical, skewed data	Can't compute on text/categories
Mean	Numerical, symmetric	Requires arithmetic; invalid for non-numbers

MEASURES OF POSITION

Measure of position identifies the position of a value, relative to other values in a set of data. The most common measures of position are Quartile, Decile, Percentile.

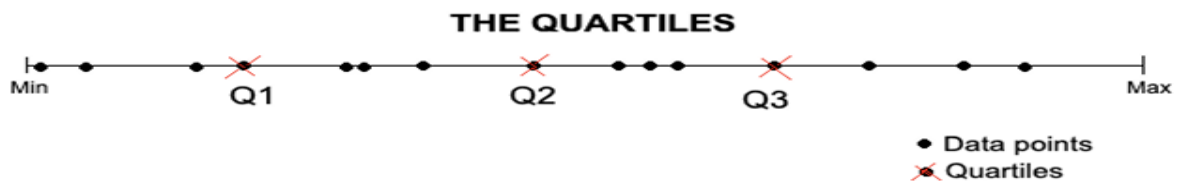
QUANTILES

Quantiles are statistical values that divide an ordered data set into equal-sized, continuous intervals. They help us understand the distribution of data by indicating the points below which a certain fraction of the data falls.

The measures that depend up on the **positions in distribution are quartiles (4 parts), deciles (10 parts), and percentiles (100 parts)**, collectively called **Quantiles**.

Quartiles:

Quartiles are measures that divide the frequency distribution into four equal parts.



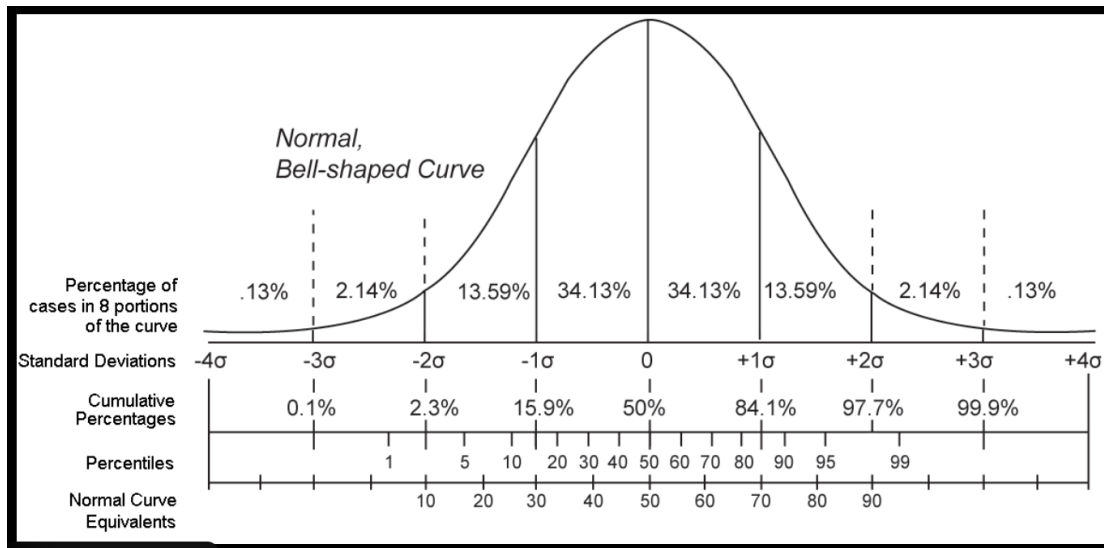
The value of the variables corresponding to these divisions are denoted Q1, Q2, and Q3 often called the first, the second and the third quartile respectively.

Q1 is a value which has 25% items which are less than or equal to it. Similarly Q2 has 50% items with value less than or equal to it and Q3 has 75% items whose values are less than or equal to it.

Deciles divides the data in 10 equal parts.

Percentile divides the data in 100 equal parts

For any data $Q2 = D5 = P50 = \text{Median}$



While computing these values in discrete data, we add 1 to N whereas in continuous series we do not add 1

To find Q1, Q3.....

- Arrange the values in ascending order.

- **For raw data**

$$Q1 = \frac{1}{4}(n+1)^{th} \text{ value}$$

$$D6 = \frac{6}{10}(n+1)\text{th term}$$

$$Q3 = \frac{3}{4}(n+1)^{th} \text{ value}$$

$$P20 = \frac{20}{100}(n+1)\text{th term}$$

- **For Discrete data** : find cumulative frequency and find the quantile with (n+1) terms. The value corresponding to the cf, just greater than the above value is required result.

- **For Continuous data** calculate the cumulative frequencies corresponding to each value.

$$Q1 = l + \left(\frac{\frac{n}{4} - cf}{f} \right) \times h$$

l = lower limit of class containing the score

n = total number of values

€

cf = cumulative frequency of preceding class

f = frequency of class interval

h = class size

$$Q3 = l + \left(\frac{\frac{3n}{4} - cf}{f} \right) \times h$$

Ex: Find Q1, Q2, Q3 for the data:

5, 7, 9, 12, 15, 18, 20, 22, 25

$$Q_1 = \frac{1(9+1)}{4} = \frac{10}{4} = 2.5^{th} item$$

(between 2nd term and 3rd term)

$$\text{Therefore } Q_1 = 7 + 0.5(9 - 7)$$

$$Q_2 = 2(9+1)/4 = 5^{th} \text{ term} = 15$$

$$Q_3 = 3(9+1)/4 = 7.5^{th} \text{ term} = 7 + 0.5(22 - 20) = 21$$

Ex: Find Q1, Q2, Q3 for given data

x	f	CF
10	3	3
15	5	8
20	7	15
25	3	18
30	2	20

$$Q_1 = (20 + 1)/4 \text{ term} = 5.25^{th} \text{ term} = 15$$

$$Q_2 = (20+1)/2 \text{ th term} = 10.5^{th} \text{ term} = 20$$

$$Q_3 = 3(20+1)/4^{th} \text{ term} = 15.75^{th} \text{ term} = 25$$

Ex: Find Q1, Q2, Q3 for given data

Class	f	CF
0-10	5	5
10-20	9	14
20-30	14	28
30-40	8	36
40-50	4	40

$$Q_1 = 40/4^{th} \text{ term} = 10^{th} \text{ term (10-20)}$$

$$Q_1 = 10 + \frac{(10 - 5)}{9} \times 10$$
$$= 10 + 5.56 = 15.56$$

Pandas quantile(q) computes the value at fraction q (0-1); use lists for multiples.
NumPy np.quantile(array, q) works similarly for arrays.

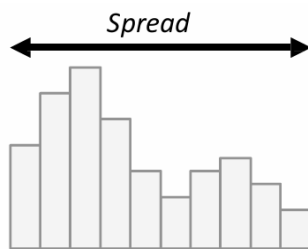
MEASURE OF SPREAD

Dispersion:

Dispersion in statistics refers to the degree to which the data tends to spread around the central value (mean or median) is called **Dispersion or Variation of the data**.

It indicates the variability or spread of data, helping to understand how consistent or diverse the data values are. If the spread or dispersion is less, the **data is preferred for Machine Learning**.

It should not be unduly affected by outliers (extreme values).



Measures of Dispersion / Variation in a distribution

Absolute measures: These measures of dispersion are expressed in the unit of variable itself like Kg, cm, marks etc. They are **Range, Interquartile Range, Quartile Deviation, Mean deviation**

Relative measures: These are also known as **Coefficient of Dispersion**, obtained as ratio or percentage of the average and are **independent of the units**.

- **Coefficient of range**
- **Coefficient of mean deviation**

The dispersion in a data is measured on the basis of the observations and the types of the measure of central tendency used. Several measures of Dispersion are there among which the common are:

- Range (R)
- Quartile Deviation (QD)
- Mean Deviation (MD)
- Standard Deviation (SD)

RANGE:

Range is the difference between the highest and lowest value of a given distribution. It provides a quick estimate of the variability distribution. Range is calculated based upon two extreme values only. It gives a crude measure of spread but **is sensitive to outliers**.

Until max. , min., value remain intact, any change in other values will not affect the range.

$$\text{Range} = \text{Maximum value} - \text{Minimum value}$$

For continuous data:

- Find difference between **lower limit of lowest class and upper limit of highest class**
- Find difference between **mid-points of the highest class and the lowest class**.

Ex. Find the range of the salaries of the staff of a school. (ans: 75000)

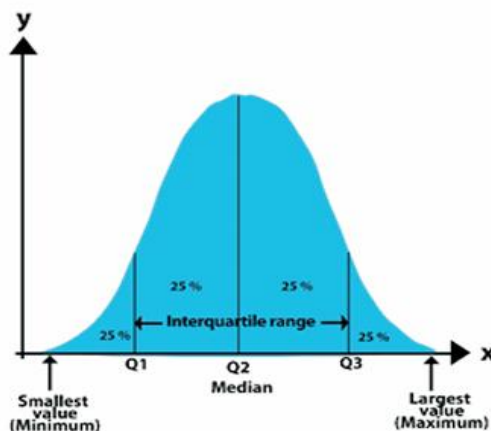
Staff	Salary (Rs.)
Principal:	100000
Senior Teacher	70000
Junior Teacher	45000
Other workers	25000

$$\text{Coefficient of range} = \frac{\text{Max.} - \text{Min.}}{\text{Max.} + \text{Min.}}$$

If the average of two distributions are about the same, then a comparison of range indicates that the distribution with smaller range has less dispersion.

QUARTILE DEVIATION: (Semi Inter Quartile Range)

Quartiles separate the original set of data into four equal parts. Each of these parts contain one-quarter (25%) of the data where two halves are less than median and other two greater than median.



- The difference between the third and first quartile is called the **Interquartile Range**

$$\text{IQR} = Q_3 - Q_1$$
- The IQR represents the spread of the middle 50% of the data and is called the middle half and is effective in identifying the outliers in data.
- An outlier is any value at least 1.5 times IQR, above Q_3 or below Q_1 .**

$$\text{Quartile deviation (Q.D.)} = \frac{Q_3 - Q_1}{2} \quad (\text{semi inter quartile range})$$

$$\text{Coefficient of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

To find $Q_1, Q_3, \dots$

- Arrange the values in ascending order.
- For ungrouped data

$$Q_1 = \frac{1}{4}(n+1)^{\text{th}} \text{ value}$$

$$Q_3 = \frac{3}{4}(n+1)^{\text{th}} \text{ value}$$
- For grouped data calculate the cumulative frequencies corresponding to each value.

$$Q_1 = l + \left(\frac{\frac{n}{4} - cf}{f} \right) \times h$$

l = lower limit of class containing the score

n = total number of values

cf = cumulative frequency of preceding class

f = frequency of class interval

h = class size

$$Q_3 = l + \left(\frac{\frac{3n}{4} - cf}{f} \right) \times h$$

Remember: For individual and discrete data add 1 to N , and for continuous data do not add 1.

So, $Q_1 = [(N+1)/4]^{\text{th}}$ term for individual and $(N/4)^{\text{th}}$ term for continuous data.

$Q_3 = [3(N+1)/4]^{\text{th}}$ term for individual and $(3N/4)^{\text{th}}$ term for continuous data.

$D_1 = [(N+1)/10]^{\text{th}}$ term for individual and $(N/10)^{\text{th}}$ term for continuous data.

$D_6 = [6(N+1)/10]^{\text{th}}$ term for individual and $(6N/10)^{\text{th}}$ term for continuous data.

$P_{30} = [30(N+1)/100]^{\text{th}}$ term for individual and $(30N/100)^{\text{th}}$ term for continuous data.

$P_{90} = [90(N+1)/100]^{\text{th}}$ term for individual and $(90N/100)^{\text{th}}$ term for continuous data.

Ques : For the given data 3, 5, 7, 9, 15, 18, 20, 30

Calculate mean, median, mode, Q_1 , Q_3 , IQR, Range, Coeff of QD.

Mean = 13.375

Median = 12

Mode = no mode

$$Q1 = \left(\frac{8+1}{4} \right)^{th} = \left(\frac{9}{4} \right)^{th} = 2.25^{th} \text{ term}$$

$$Q1 = 2^{nd} \text{ term} + 0.25 \times (3^{rd} - 2^{nd}) = 5 + 0.25 \times (7 - 5) = 5 + 0.5 = 5.5$$

$$Q3 = \left(\frac{3(8+1)}{4} \right)^{th} = \left(\frac{27}{4} \right)^{th} = 6.75^{th} \text{ term}$$

$$Q3 = 6^{th} \text{ term} + 0.75 \times (7^{th} - 6^{th}) = 18 + 0.75 \times (20 - 18) = 18 + 1.5 = 19.5$$

When position is a **whole number**, directly pick that term — no interpolation needed.

For the given data, find the mean, median, mode, Q1, Q3, IQR, Range, Coeff of QD:

X	0	1	2	3	4	5
Freq	2	4	3	6	3	2

X	f	fx	cf
0	2	0	2
1	4	4	6
2	3	6	9
3	6	18	15
4	3	12	18
5	2	10	20
20		50	

Q1 = 1, Q2 = 3, Q3 = 4

QD = Q3 - Q1 = 4 - 1 = 3

Semi QD = 3/2 = 1.5

Coeff of QD = $\frac{Q3 - Q1}{Q3 + Q1} = \frac{3}{5} = 0.6$

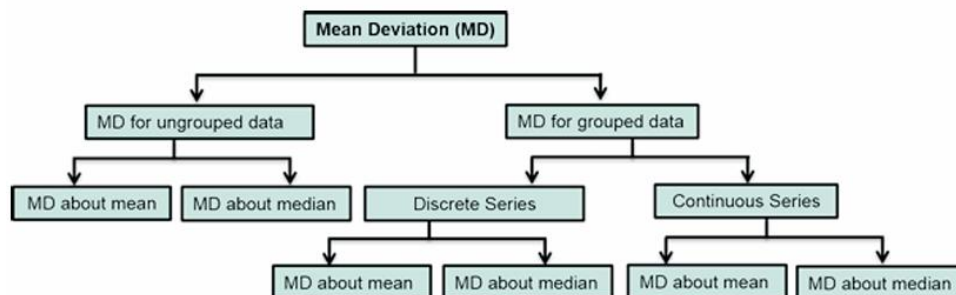
mean = $\frac{\sum fx}{N} = \frac{50}{20} = 2.5$

median = $\frac{(N+1)}{2} = \frac{21}{2} = 10.5 = 3$ (Q2)

Q1 = $\left(\frac{N+1}{4} \right) = \frac{21}{4} = 5.25 = 1$

Q3 = $\left(\frac{3(N+1)}{4} \right) = \frac{3 \times 21}{4} = 15.75 = 4$

MEAN DEVIATION:



Mean Deviation for ungrouped data:

Let n observations be $X_1, X_2, X_3, X_4, X_5, \dots, X_n$ then

(I) Mean deviation about mean

$$\text{M.D. } (\bar{x}) = \frac{1}{N} \sum_{i=1}^n |x_i - \bar{x}| \quad \text{where } \bar{x} = \text{mean}$$

(II) Mean deviation about median

$$\text{M.D. } (median) = \frac{1}{N} \sum_{i=1}^n |x_i - M| \quad \text{where } M = \text{median}$$

Ex. Find the mean deviation about the mean and median for the data 4, 3, 2, 5, 7, 6, 8

(ans: 1.71, 1.71)

Mean deviation for discrete / continuous frequency distribution:

Mean deviation about mean-

$$\text{M.D. } (\bar{x}) = \frac{1}{N} \sum_{i=1}^n f_i |x_i - \bar{x}| \quad \text{where } \bar{x} = \text{mean}$$

Mean deviation about Median-

$$\text{M.D. } (median) = \frac{1}{N} \sum_{i=1}^n f_i |x_i - M|$$

$$\text{where } M (\text{Median}) = l + \left(\frac{\frac{n}{2} - cf}{f} \right) \times h$$

Ex: Find out the mean deviation about the median for following data. (ans: 1.86)

Height(cm)	147	148	150	152	155
No. of students	8	12	15	10	5

Ex: Find the mean deviation about median of following data.

Class interval	20-30	30-40	40-60	60-80	80-90
Frequency	5	10	20	9	6

Class interval (CI)	Frequency (f_i)	Mid Points (x_i)	Cf	$ x_i - M $	$f_i x_i - M $
20-30	5	25	5	25	125
30-40	10	35	15	15	150
40-60	20	50	35	0	0
60-80	9	70	44	20	180
80-90	6	85	50	35	210
	$\sum_{i=1}^7 f_i = 50$				665

$$\text{Median} = l + \left(\frac{\frac{n}{2} - cf}{f} \right) \times h = 40 + \frac{(25 - 15)}{20} \times 20 = 50$$

$$\text{M.D. (Median)} = \frac{1}{N} \sum_{i=1}^n f_i |x_i - M| = \frac{665}{50} = 13.3$$

$$\text{Coefficient of MD} = \frac{\text{mean deviation}}{\text{mean}}$$

VARIANCE:

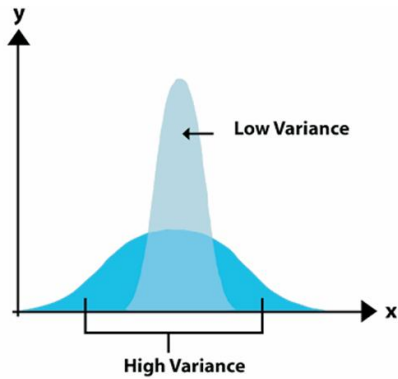
The Variance is the average of the squares of the deviation of each value from mean. It quantifies variability but is in squared units of data and is denoted by ' σ^2 ' (sigma square). So variance of n observations $x_1, x_2, x_3, \dots, x_n$ is given by:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^n (x_i - \bar{x})^2 \quad \text{Variance} = \frac{\sum (x - \bar{x})^2}{N}$$

The measures of dispersion discussed so far (range, mean deviation, quartile deviation) describes only measure of theoretical averages but does not tell anything about the spread of the distribution.

The Variance combines all the values in a data set to produce a measure of the spread.

Variance is the square of the Standard Deviation.



Ex: Find the Variance of the dataset 10, 60, 50, 30, 40, 20.

Solution: The mean of data $(\bar{x}) = \frac{10+60+50+30+40+20}{6} = \frac{210}{6} = 35$

$$\text{Variance } (\sigma^2) = \frac{1}{N} \sum_{i=1}^n (x_i - \bar{x})^2$$

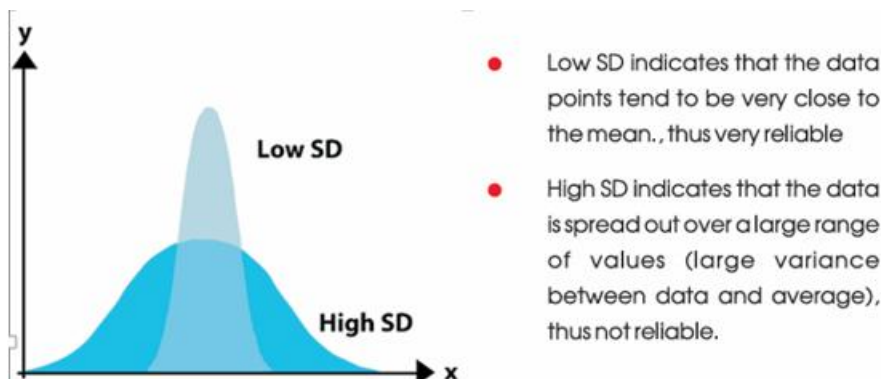
$$= \frac{(10-35)^2 + (60-35)^2 + (50-35)^2 + (30-35)^2 + (40-35)^2 + (20-35)^2}{6}$$

$$= \frac{1750}{6} = 291.7$$

STANDARD DEVIATION:

Standard Deviation (SD) measures **Variation (spread) in the data. It is the square root of variance.** It gives an idea of the shape of the distribution and indicates how much variation is there from the mean. It has wide range of application in:

- Academic Studies (coefficient of variation, hypothesis testing, Confidence Intervals)
 - Business (Variability of delivery times, inventory etc)
 - Finance (variability of returns in different asset classes)
- **Small standard deviation** → values are close to the mean (low variability)
 - **Large standard deviation** → values are spread out (high variability)



The standard deviation is (σ) =

$$= \sqrt{\frac{1}{N} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Relations between the Deviations:

$$SD = (3/2) * QD$$

$$SD = (5/4) * MD$$

Ex: Find the standard deviation of dataset 9, 3, 8, 8, 9, 8, 9, 18. (ans: 3.87)

Example 38: Thirty students were asked that how many days they require to revise the chapter of descriptive statistics. Their responses were: 4, 5, 6, 5, 3, 2, 8, 0, 4, 6, 7, 8, 4, 5, 7, 9, 8, 6, 7, 5, 5, 4, 2, 1, 9, 3, 3, 4, 6, 4. Find out the standard deviation for this data.

No. of Days (x_i)	Frequency (f_i)	($f_i x_i$)	($x_i - \bar{x}$)	($x_i - \bar{x}$) ²	$f_i (x_i - \bar{x})^2$
0	1	0	-5	25	25
1	1	1	-4	16	16
2	2	4	-3	9	18
3	3	9	-2	4	12
4	6	24	-1	1	6
5	5	25	0	0	0
6	4	24	1	1	4
7	3	21	2	4	12
8	3	24	3	9	27
9	2	18	4	16	32
	$\sum f_i = 30$	$\sum f_i x_i = 150$			$\sum f_i (x_i - \bar{x})^2 = 152$

means $\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{150}{30} = 5$

Standard Deviation (σ) = $\sqrt{\frac{1}{N} \sum_{i=1}^n f_i (x_i - \bar{x})^2} = \sqrt{\frac{152}{30}} = 2.25$

Example 39: In a test of 50 students, marks secured by students is given in the following table. Calculate the standard deviation of scores.

Height (inch)	10-20	20-30	30-40	30-40	40-50
Number of students	5	8	15	16	6

Ans: 11.48

COEFFICIENT OF VARIATION

It is a standardized measure of dispersion calculated as the ratio of standard deviation to the mean, often expressed as %

$$CV = \frac{\sigma}{\mu} \times 100$$

Coefficient of variation is a dimensionless quantity, making it useful for comparing dispersion between datasets with different units.

MEASURE OF ASYMMETRY

NEED OF MEASURE OF ASYMMETRY

Two distribution may have same mean and Standard Deviation but may differ widely in their overall appearance.

$$\bar{x} = \frac{\sum fm}{N}, N = \sum f, \sigma = \sqrt{\frac{\sum f(m - \bar{x})^2}{N}}$$

X	Y
0-5	10
5-10	<u>30</u>
10-15	60
15-20	<u>60</u>
20-25	<u>30</u>
25-30	10

N = 200

Mean = 15

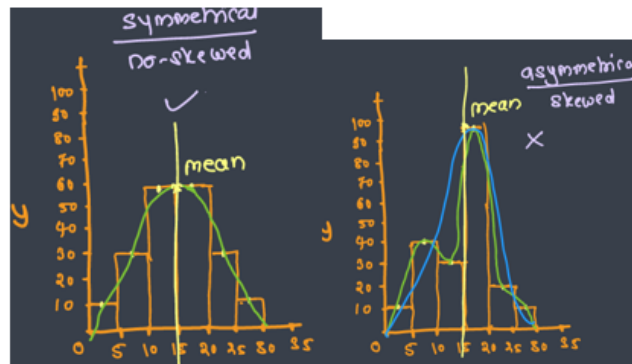
S.D. = 6.02

X	Y
0-5	10
5-10	<u>40</u>
10-15	<u>30</u>
15-20	<u>90</u>
20-25	<u>20</u>
25-30	10

N = 200

mean = 15

S.D. = 6.02



SKEWNESS:

Skewness denotes the degree of asymmetry in the data i.e. it **measures the deviation of a random variables** for the given distribution from the normal distribution, which is symmetrical on both sides. Measure of skewness tell us the direction and the extent of skewness.

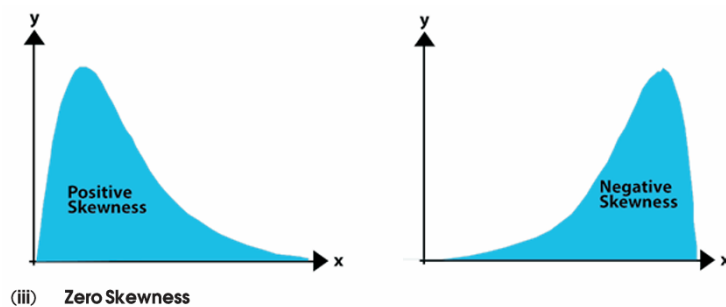
Skewness can be positive, negative, or zero. $-1 \leq \text{skewness} \leq +1$

Positive Skewness

If the distribution is shifted to the **left with its tail on the right side, it is positively skewed**. It is also called the right-skewed distribution. A positive skewed distribution assumes a skewness value of more than 0. In this case, distribution has long tail on right and Mean > Median > Mode.

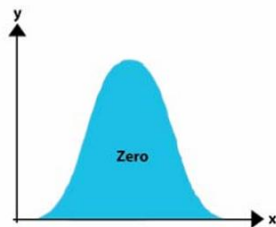
Negative Skewness

If the distribution is shifted to the **right with its tail on the left side, it is negatively skewed**. It is also called as left- skewed distribution. A negative skewed distribution assumes a skewness value of less than 0. In this case, distribution has long tail on left and Mean < Median < Mode.



Zero Skewness

When skewness is zero, distribution is called 'Symmetrical'



DISPERSION vs SKEWNESS

- Dispersion is concerned with the amount of variation rather than direction.
- Skewness tells the direction of variation from symmetry

TESTS OF SKEWNESS

In order to assert if a distribution is skewed or not, the following tests may be applied.

- The values of mean, mode, median do not coincide
- The data plotted on a graph should not give the normal bell-shaped curve.
- Quartiles are not equidistant from the median.

MEASURES OF SKEWNESS

Absolute Measure

- It is the difference between the mean and median. (mean – median)
- If value of mean is greater than median, the skewness will be positive.
- If value of median is greater than mean, the skewness will be negative.

Relative Measure

- **Karl Pearson's coefficient of skewness**

*It is known as pearsonian coefficient of skewness, whose value lies between +1 or -1.

*When distribution is symmetrical mean, mode, median will coincide and result is 0.

*The coefficient will be +ive sign for positive skewness and –ive sign for negative skewness.

The image shows two handwritten formulas for Karl Pearson's coefficient of skewness (Sk_p) on a dark background with yellow boxes. The left box is titled "using mode" and contains the formula $Sk_p = \frac{\text{mean} - \text{mode}}{\sigma}$. The right box is titled "using median" and contains the formula $Sk_p = \frac{3(\text{mean} - \text{median})}{\sigma}$.

- **Bowley's coefficient of skewness (Quartile coefficient of skewness)**

*It is based on Quartiles

*In symmetrical distribution, first and third quartiles are equidistant from the median.

*If distribution is positively skewed, top 25% of values will tend to be farther from median than the bottom 25% values.i.e. Q3 will be farther from median than Q1is, and reverse for negatively skewed distribution.

$$Sk_B = \frac{Q3 + Q1 - 2 * median}{Q3 - Q1}$$

- **Kelly's coefficient of skewness**

*It can be used with any decile or percentile equidistant from median.

*It can be expressed by taking any two deciles or percentiles equidistant from median.

$$S_K = \frac{D1 + D9 - 2 * median}{D9 - D1}$$

Decile coeff of skewness

$$S_K = \frac{P10 + P90 - 2 * median}{P90 - P10}$$

Percentile coeff of skewness

Measure of skewness based on moments

Measure of skewness: Coefficient of skewness

- (i) Pearson's 1st coefficient of skewness $= \frac{Mean - Mode}{S.D.}$
- (ii) Pearson's 2nd coefficient of skewness $= \frac{3(Mean - Median)}{S.D.}$
- (iii) Quartile coefficient of skewness $= \frac{Q_3 + Q_1 - 2Q_2}{S.D.}$
- (iv) Percentile coefficient of skewness $= \frac{P_{90} + P_{10} - 2Median}{P_{90} - P_{10}}$
- (v) Decile coefficient of skewness $= \frac{D_9 + D_1 - 2Median}{D_9 - D_1}$

KURTOSIS

Kurtosis is a statistical measure that describes the **shape of a distribution**, specifically how **peaked or flat** it is as compared to a normal distribution. So It focuses on the **tails and peak** of the data distribution.

Kurtosis is a measure of the **tailedness** of a distribution, i.e.how often outliers occur, and degree of '**peakedness**' of frequency curve.

β_2 is the **coefficient of kurtosis**, a numerical measure used to describe the **shape of a distribution**, especially its **peakedness and tail thickness**. It is **always positive** (because it's a ratio of squared quantities).

But in practice, we often use “**excess kurtosis**”, which **can be positive, zero, or negative**.

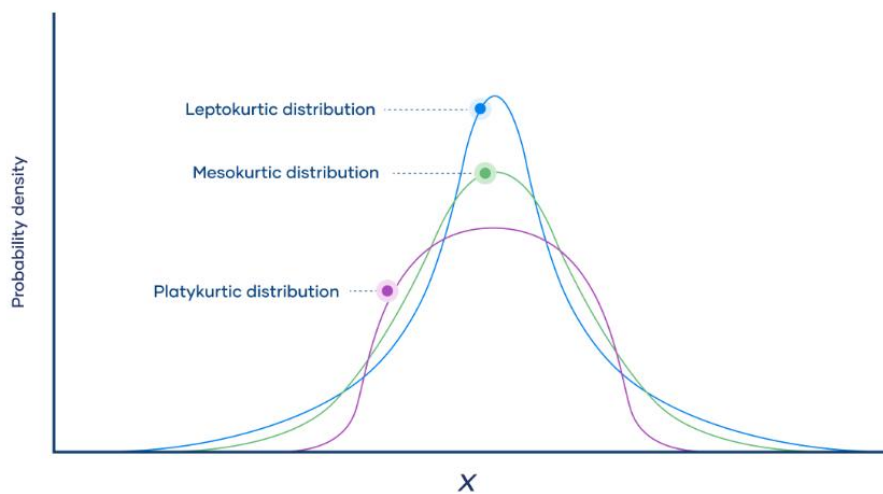
$$\text{Excess Kurtosis} = \beta_2 - 3$$

For a **normal distribution** $\beta_2 = 3$, So we subtract 3 to make the normal case **zero** for easy comparison.

Types of kurtosis:

1. **Leptokurtic ($\beta_2 > 3$)** (positive kurtosis [excess kurtosis > 0])
 - More peaked than normal
 - Heavy tails (more outliers)
2. **Mesokurtic ($\beta_2 = 3$)** (zero kurtosis [excess kurtosis $= 0$])
 - Normal distribution shape
 - Moderate tails
3. **Platykurtic ($\beta_2 < 3$)** (negative kurtosis [excess kurtosis < 0])
 - Flatter peak than normal
 - Light tails (fewer outliers)

Tails are the tapering ends on either side of a distribution. Tails represent how often **outliers** occur.



	Category		
	Mesokurtic	Platykurtic	Leptokurtic
Tailedness	Medium-tailed	Thin-tailed	Fat-tailed
Outlier frequency	Medium	Low	High
Kurtosis	Moderate (3)	Low (< 3)	High (> 3)
Excess kurtosis	0	Negative	Positive
Example distribution	Normal	Uniform	Laplace

Golden Rule: When data has outliers → prefer **Median, IQR, and Coeff of QD** over Mean and Range. When data is clean and symmetric → **Mean and Range** work well.